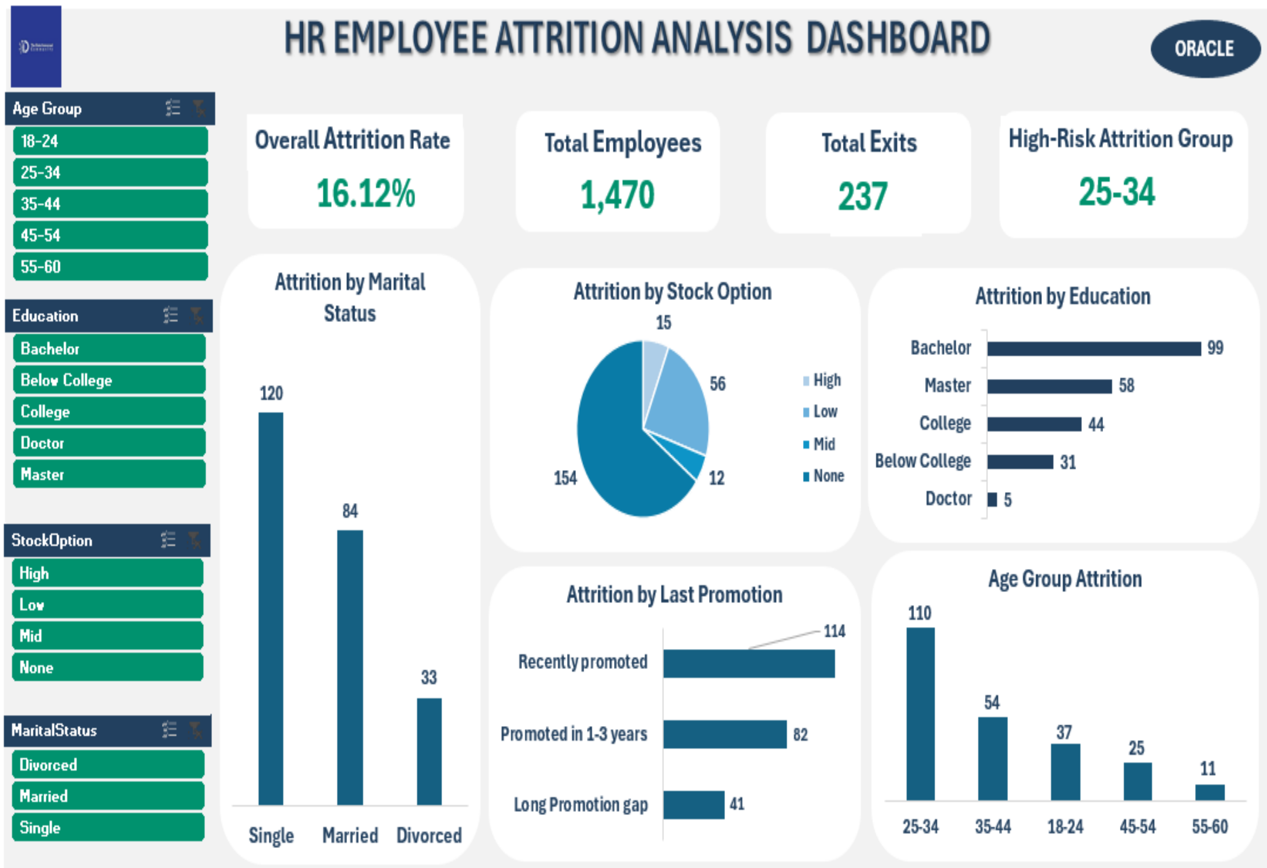
HR Employee Attrition Analytics

ABSTRACT



HR EMPLOYEE ATTRITION ANALYSIS DASHBOARD
ORACLE
Age Group
18-24
25-34
35-44
45-54
55-60
Education
Bachelor
Below College
College
Doctor
Master
StockOption
High
Low
Mid
None
MaritalStatus
Divorced
Married
Single
Overall Attrition Rate
16.12%
Total Employees
1,470
Total Exits
237
High-Risk Attrition Group
25-34
Attrition by Marital Status
120
84
33
Single
Married
Divorced
Attrition by Stock Option
15
56
154
12
High
Low
Mid
None
Attrition by Education
Bachelor 99
Master 58
College 44
Below College 31
Doctor 5
Attrition by Last Promotion
Recently promoted 114
Promoted in 1-3 years 82
Long Promotion gap 41
Age Group Attrition
110
54
37
25
11
25-34
35-44
18-24
45-54
55-60

INTRODUCTION

Background

This analysis has a simple but yet important concern which is: employees are leaving and TDI wants to know why. This is bound to happen in an organization but when it becomes frequent, there is a need to understand the root causes of it.

Essentially, this analysis provides clarity. It would help the management make informed and data driven decision that can improve employee experience and retention, creating a stable work environment,

Problem Statement

The core problem of this analysis is to uncover the reasons why employees are leaving the firm and what are the key factors across the dataset contributing to this attrition.

Research Questions

1. Does Employee age contribute to attrition?

2. How does marital status influence attrition?

3. Is there a relationship between level of education and attrition?

4. Is stock option a major driver of attrition?

5. Does year since last promotion affect attrition?

Objectives

1. To determine whether certain age groups have higher attrition rates.

2. To evaluate how marital status influences attrition.

3. To compare the different levels of education and attrition.

4. To examine whether employees with less stock option are more likely to leave.

5. To ascertain if the year since last promotion is a trigger for employees to leave.

DATA DESCRIPTION

Data Sources: The dataset is from an internal data source which is the TDI organization.

Data Characteristics: The dataset is made up of 7 variables consisting of integers, Boolean and character data types.

METHODOLOGY

Data Cleaning

1. Duplicate Removal: In my data cleaning procedure, I checked for duplicates and found no duplicate entry in the dataset.

2. Blank Spaces: Using the Go to special Excel tool, I examined my data for blank spaces or cells and found none.

3. Filter: I used the filter function to sort my data, check for inconsistencies and reviewed specific categories of my variables.

4. Data Grouping: While cleaning my data set, I grouped the age column into different age groups (18-24, 25-34, 35-44, 45-54, 55-60) and the years since last promotion (Recently promoted, Promoted in 1-3 years and Long Promotion gap) using Excel IF function.

I also converted the attrition column to a numeric format (1,0) using Excel function allowing for attrition rate calculation and average use in my pivot table.

5. **Conversion of Data to Number Format**: During my data cleaning process, I converted the Employee ID and Age column to number format to ensure all values are treated as numeric data rather than text.

Exploratory Data Analysis

Key Performance Indicators

For my analysis, I had the following Key Performance Indicators

a. Overall Attrition Rate: This measures the percentage of employees who left the organization.

b. Total Employees: This measures the total size of the organization workforce.

c. Total Exits: This reflects the actual number of employees who existed, that is those who left.

d. High Risk Attrition Group: This measure shows the age group with the highest attrition rate.

ANALYSIS AND FINDINGS

Analysis Questions

1. Does Employee age contribute to attrition?

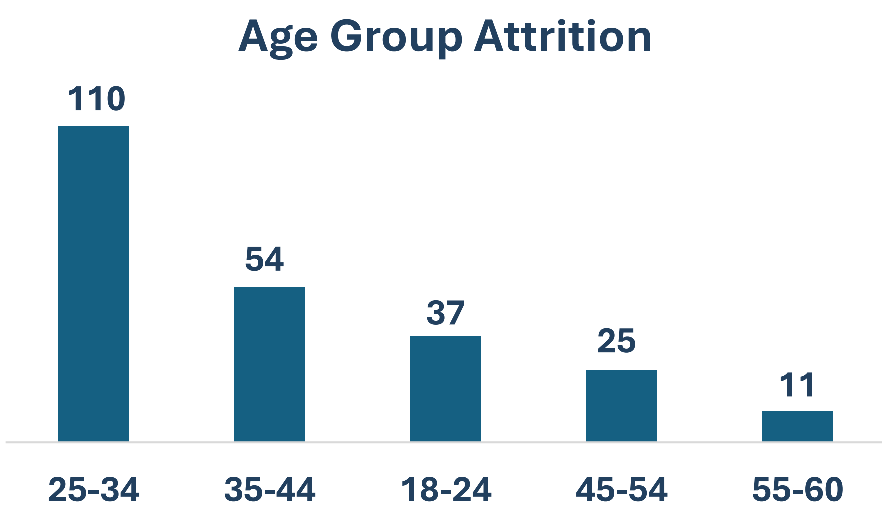


Figure 1: Age Group Attrition

From the chart above, the 25-34 age group experiences the highest level of attrition compared to other groups. This shows that the younger employees are more likely to leave the company.

2. How does marital status influence attrition?

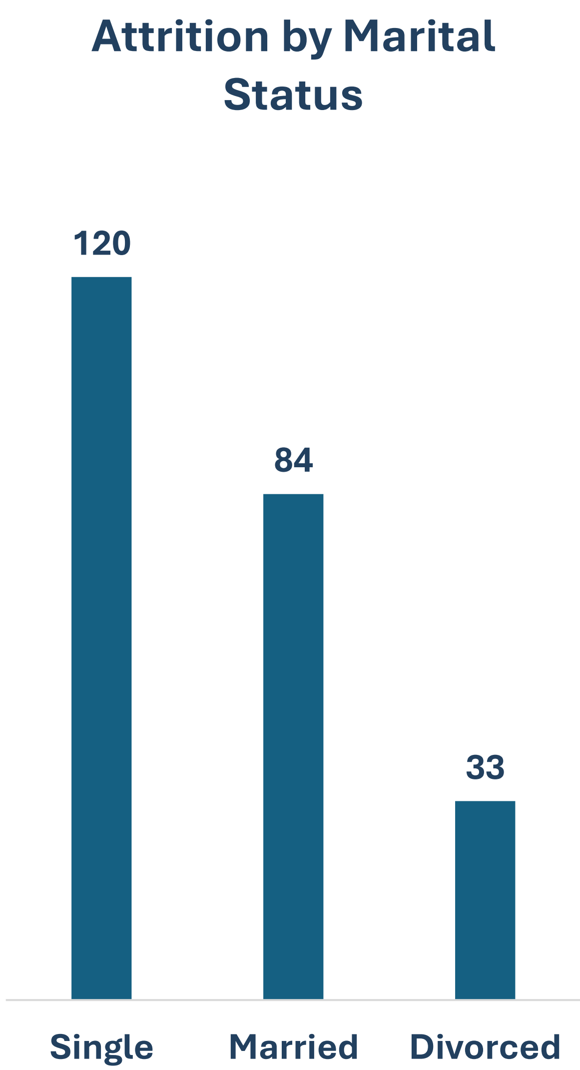


Figure 2: Attrition by Marital Status

Looking at the marital status chart, there is a clear indication that single employees have the highest attrition, followed by the married employees while divorced employees show the lowest levels of attrition in the dataset.

3. Is there a relationship between level of education and attrition?

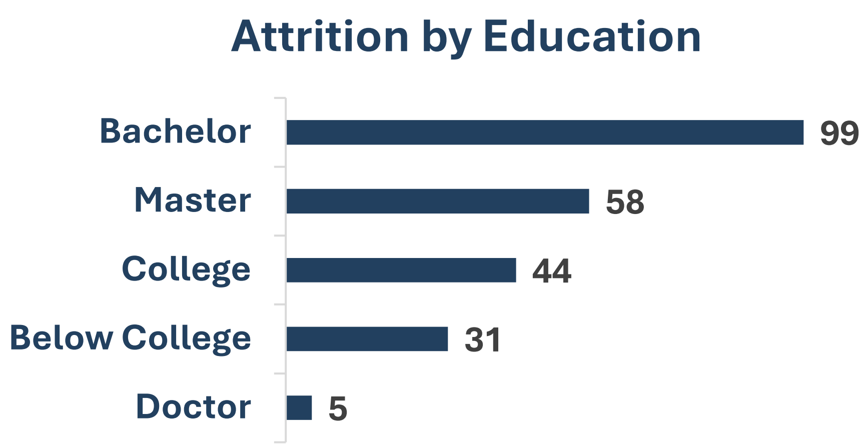


Figure 3: Attrition by Education

Education appears to play a significant role in attrition. The chart shows that employees with Bachelor's and Master's degree experience higher attrition while those with Doctorate degree show the lowest attrition levels.

4. Is stock option a major driver of attrition?

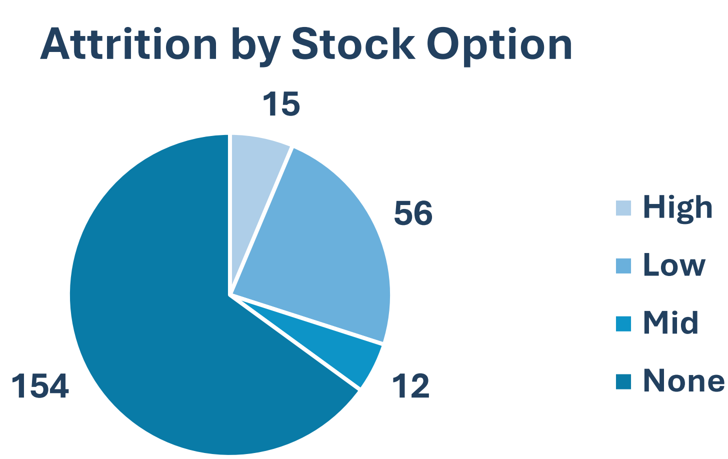


Figure 4: Attrition by Stock Option

From the chart, employees with no Stock Option show the highest attrition, while those with higher stock option benefits appear less likely to leave.

5. Does year since last promotion affect attrition?

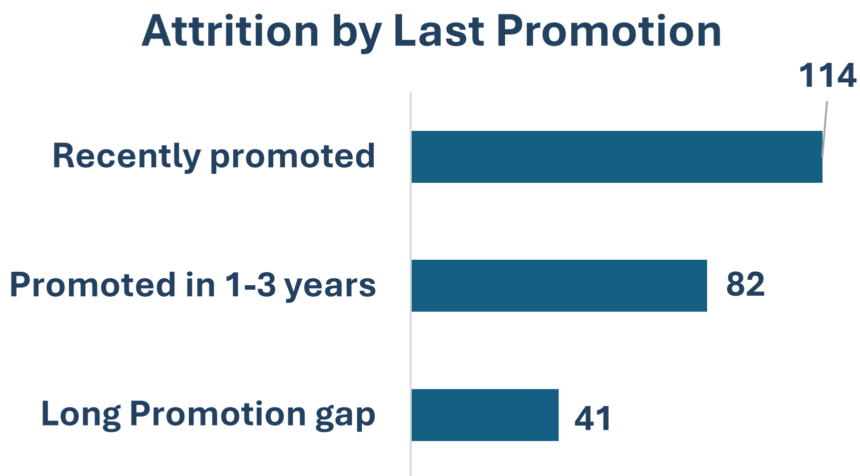


Figure 5: Attrition by Last Promotion

Finally, the analysis of attrition by last promotion shows that employees who were recently promoted have the highest attrition rate compared to those promoted 1-3 years ago or those with long promotion gap.

Key Performance Indicators

For my analysis, I had the following Key Performance Indicators

a. Overall Attrition Rate:

This KPI shows the proportion of employees who have left the organization compared to the total workforce. An attrition rate of 16.12% means that 16 out of every 100 employee in the company are no longer with the organization during the period analyzed.

b. Total Employees:

This represents the entire number of employees included in the dataset used for the analysis. Here the organization has 1,470 employees which forms the basis for calculating other indicators.

c. Total Exits:

This shows the actual number of employees who have left the company. Out of the 1,470 employees, 237 employees exited the organization.

d. High Risk Attrition Group:

This indicator highlights the group of employees that appears to be most likely to leave the organization. Employees between 25 and 34 years old have the highest attrition compared to other age groups.

RESULT

Summary of Findings

The HR Attrition analysis shows that out of 1,470 employees, 237 employees exited, resulting in an overall attrition rate of 16.12%. While most employees remain within the organization, the attrition level suggests there is still need for strategies that will improve employee retention.

The analysis indicates that younger employees 25-34 experience the highest attrition, showing that early career staff are more likely to leave for better opportunities. In terms of marital status, single employees show the highest attrition while married and divorced employees appear to be more stable. Also stock option influence retention as employees with no stock option have the highest exits. Additionally, employees who were recently promoted show a higher likelihood of leaving the organization compared to those who have experienced longer gaps.

Overall, the findings shows that age, education, marital status, financial incentive (stock option) and year since last promotion are key factors influencing employee attrition in TDI.

Visualizations

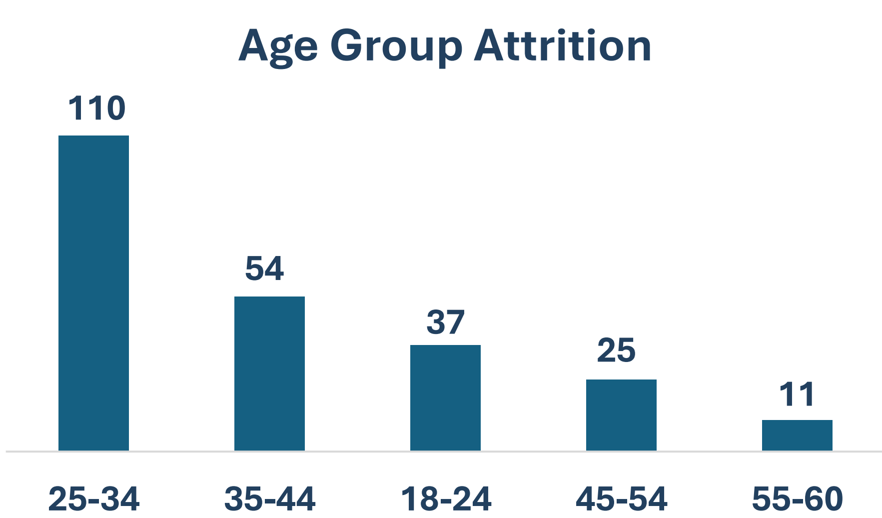


Figure 1: Age Group Attrition

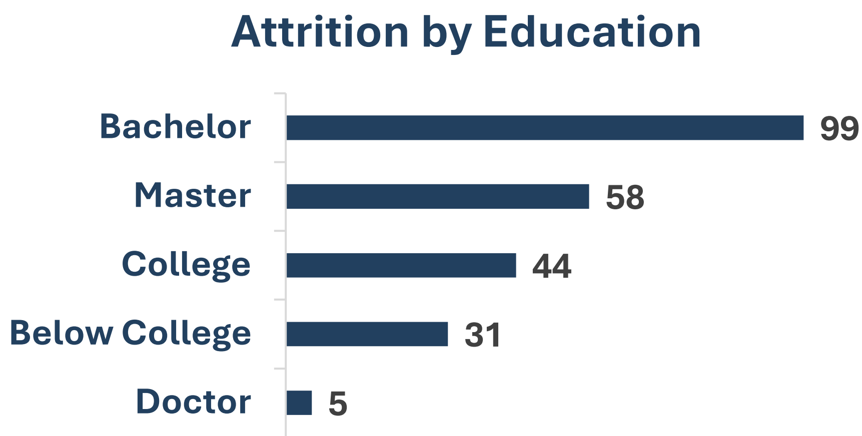


Figure 2: Attrition by Education

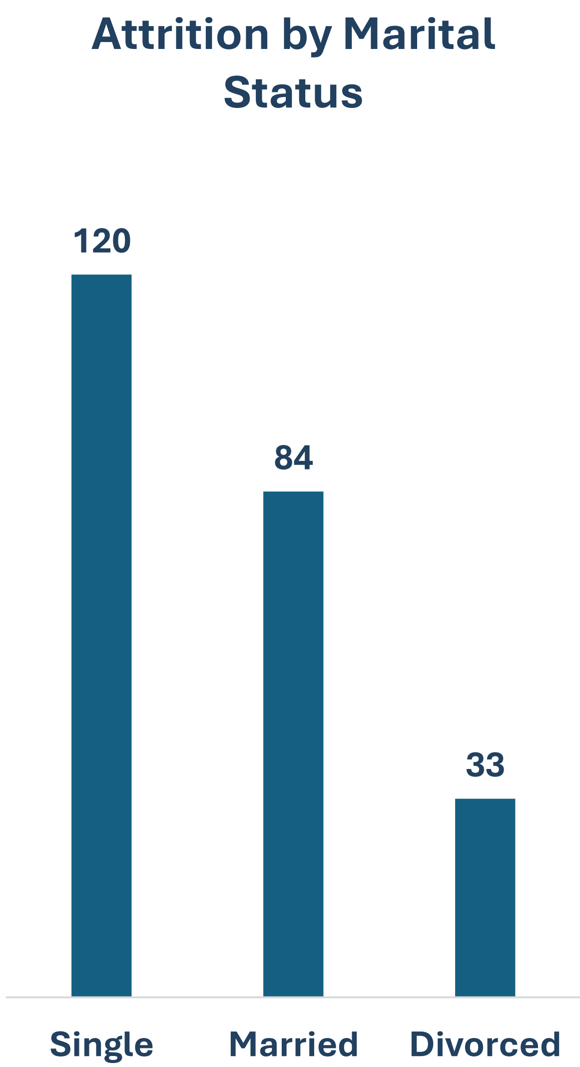


Figure 3: Attrition by Marital Status

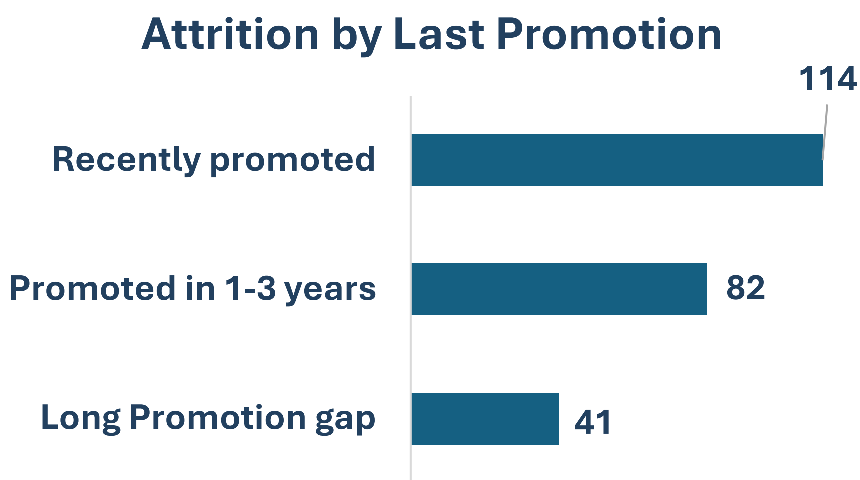


Figure 4: Attrition by Last Promotion

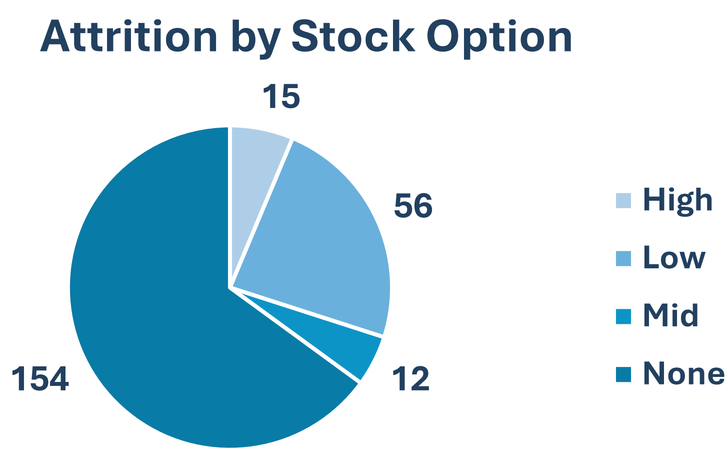
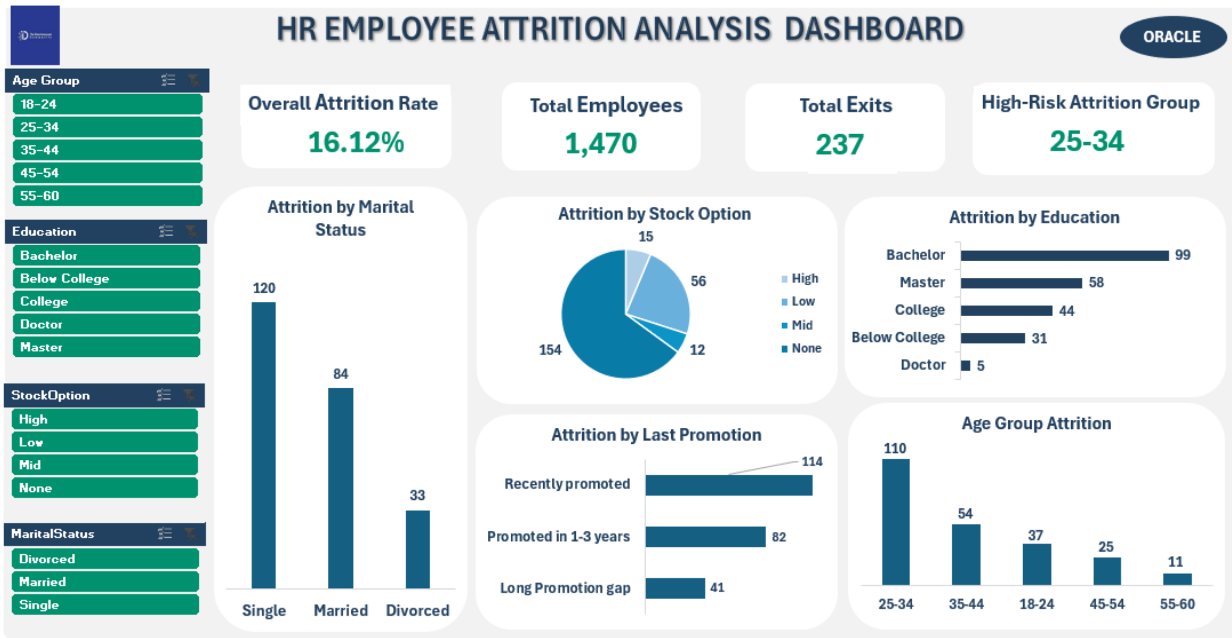


Figure 5: Attrition by Stock Option

DASHBOARD



[Click here to view dashboard]

The HR Attrition Dashboard provides a clear overview of employee turnover within the organization. Out of 1,470 employees, 237 have existed, resulting in an attrition rate of 16.12%. This gives a quick understanding of the level of workforce turnover. The dashboard also highlights important patterns. Younger employees are more likely to leave the organization. In addition, employees who do not receive stock options and those who have been recently promoted show higher attrition, indicating that incentives and career opportunities play an important role in employee retention.

Overall, the dashboard helps turn HR data into meaningful insights by clearly showing the factors that may influence employee attrition, allowing HR managers to better understand where retention efforts should be focused.

DISCUSSION

Interpretation of Results

1. Attrition by Age: Employees in the 25-34 age group are in the early stages of their careers, exploring opportunities, gaining experiences, or searching for roles that align with long-term goals. The 18-24 group also shows noticeable attrition, while older employees particularly those in the 45-54 range appear to leave less frequently. This pattern indicates that as employees grow older and settle in their careers, they tend to remain with the company longer.

2. Attrition by Marital status: Single employees generally have fewer personal or family commitments tying them to a particular location or job, making it easier for them to pursue new opportunities. Married employees have more stability due to family responsibilities which encourages them to remain longer in their roles.

3. Attrition by Education: From the results, employees with mid-level education are more likely to move between organizations in search of better career growth, while those with advanced degrees holding senior roles that provide greater stability and long-term opportunities within the organization.

4. Attrition by Stock option: Stock option serves as long-term financial incentives encouraging employees to remain with the organization since leaving means losing those benefits.

5. Attrition by Last promotion: The indicates that promotion alone does not necessarily improve employee retention. In most cases, newly promoted employees face increased expectations, role adjustments, or external opportunities that make them more likely to leave the organization shortly after promotion.

Implications

The findings show that the organization will face challenges retaining younger employees who are more likely to be single. The organization needs to invest in career development programs, mentorship, and clear advancement opportunities to retain younger employes and reduce early career attrition. Engagement programs need to be strengthened. Also, competitive compensation and reward systems play an important role in employee retention.

Based on education, there is a need for tailored career development programs, training opportunities, and growth pathways to meet the needs of employees with different educational qualifications.

Lastly, promotion does not guarantee retention. Organizations need to introduce role transition guidance to help employees adjust to their new responsibilities

CONCLUSION

Summary

My Analysis examined employee attrition across key demographic and job-related factors such as age group, marital status, education level, stock option, and years since last promotion. The results highlight noticeable patterns showing that turnover is more common among younger employees, single staff, employees without stock options and those who have been recently promoted.

Overall, the analysis shows that while TDI maintains a relatively stable workforce, certain employee groups are more likely to leave. By focusing on career growth opportunities, employee incentives and engagement strategies, the organization can better manage attrition and improve long-term workforce stability.

Recommendations

In order to minimize attrition rate TDI should take these practical steps;

1. Provide stronger support for younger employees: Since employees in the 18-24 age group appear to leave, the organization should focus on helping early-career staff settle into the workplace. Mentorship programs and training opportunities will help younger employees feel more confident about building their future within the organization.

2. Retention incentives for high performers: Introduce incentives such as recognition programs, development opportunities, or compensation adjustments to retain promoted talent.

3. Strengthen incentive programs: Employees who do not receive stock options have higher attrition. Expanding financial incentives, recognition programs, or performance-based rewards will encourage employees to remain with the organization for longer.

4. Focus on retaining high-risk groups: Some employee groups appear more likely to leave than others, HR should focus more attention on these groups by introducing retention strategies such as training, development programs and improved employee support system.